

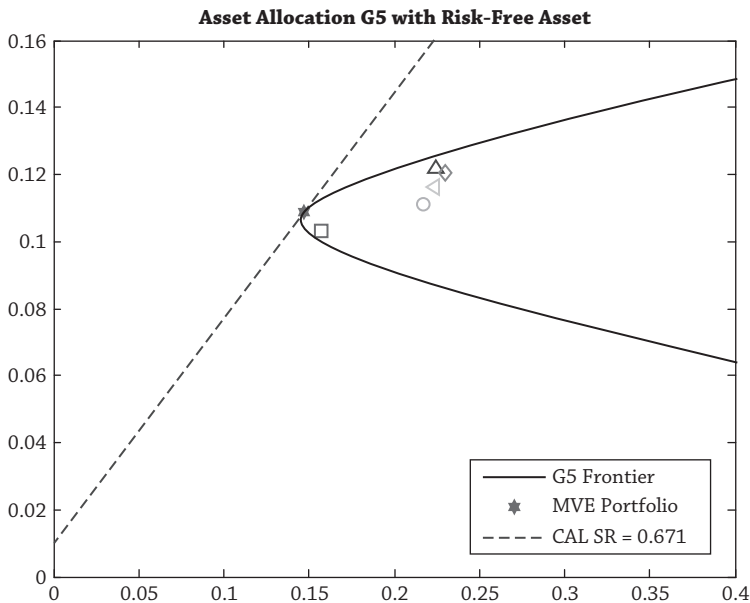


Figure 3.10

taking all combinations of the MVE with the 1% yielding risk-free asset. The MVE itself corresponds to a 100% position in only G5 equities and the intersection point of the CAL on the y-axis at 1% corresponds to a 100% risk-free position.

The slope of the CAL represents the portfolio’s Sharpe ratio. Since the CAL is tangent at the MVE, it represents the maximum Sharpe ratio that can be obtained by the investor. A line that starts from the risk-free rate of 1% on the y-axis but with a larger angle, which tilts closer to the y-axis, cannot be implemented as it does not intersect the frontier. The frontier represents the set of best possible portfolios of G5 risky assets, and we must lie on the frontier. A line that starts from the risk-free rate of 1% on the y-axis but with a lesser angle than the CAL, which tilts closer to the x-axis, intersects the frontier. These are CALs that can be obtained in actual portfolios but do not represent the highest possible Sharpe ratio. The maximum Sharpe ratio is the tangency point, or MVE.

The MVE in Figure 3.10 has a Sharpe ratio of 0.671. It consists of:

	U.S.	JP	U.K.	GR	FR
MVE Portfolio	0.53	0.24	0.12	0.10	0.02

All the portfolios that lie on the CAL have the same Sharpe ratio, except for the 100% risk-free position that corresponds to the risk-free rate of 1% on the y-axis.

Now that we’ve found the best risky MVE portfolio, the investor mixes the risk-free asset with the MVE portfolio. This takes us off the mean-variance frontier.